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Applications of AI/ML Algorithms in L3 Handover

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Abstract

Ensuring seamless connectivity in cellular networks is essential, particularly as mobile devices, including Internet of Things (IoT) nodes, move between the coverage areas of various base stations. Traditionally, such mobility has been managed through Layer 3 (L3) handover mechanisms, predominantly using Radio Resource Control (RRC) signaling within LTE and 5G architectures. These procedures rely on routine signal strength measurements, such as Reference Signal Received Power (RSRP) and Reference Signal Received Quality (RSRQ), and are typically triggered by threshold events such as Event A3. Although effective, L3-based handovers are often hampered by latency and the overhead associated with centralized signaling.

Recent advances in Artificial Intelligence (AI) and Machine Learning (ML) have demonstrated significant potential in refining L3 mobility management strategies. By analyzing both historical and real-time datasets, AI/ML models can anticipate user mobility patterns, optimize handover frequency, and adapt dynamically to complex and variable radio conditions. Applications include supervised learning methods such as decision trees and gradient-boosted models, with use cases in mobility prediction and RRC state transition optimization.

This thesis focuses on the application of AI/ML algorithms for L3 handover optimization. It presents a comprehensive literature review of AI/ML techniques applied to mobility management and develops practical supervised learning models - using XGBoost and Random Forest classifiers - to predict handover outcomes based on L3 mobility features. Experimental results on real mobility datasets demonstrate the effectiveness of the proposed models and provide insights into key factors influencing L3 handover performance. The work also highlights open challenges and suggests directions for future research in intelligent mobility management systems.

Acknowledgments

This thesis was conducted externally at Ericsson, and I would like to express my sincere gratitude to my industrial supervisors, senior researcher Rida Khan and experienced researcher Sahar Imtiaz for their expert guidance, insightful feedback, and steady support throughout the project. Their mentorship was instrumental in shaping the direction and outcome of this work. I am also thankful to my academic supervisor, Eunjeong Jeong at Linköping University, for her thoughtful supervision, constructive input, and ongoing encouragement during the research process. I would like to extend my appreciation to Nikolaos Pappas, my examiner at Linköping University, for his evaluation and valuable feedback, which helped strengthen the final outcome of this thesis. I am also grateful to the teams at Ericsson and Linköping University for providing a collaborative and supportive environment. Lastly, heartfelt thanks go to my family and friends for their unwavering support and motivation throughout this journey.

Contents

Abstract iii

Acknowledgments iv

Contents v

List of Figures vii

List of Tables viii

1 Introduction 1
- 1.1 Motivation . . . 1
- 1.2 Aim . . . 1
- 1.3 Research questions . . . 2
- 1.4 Delimitations . . . 2

2 Theoretical Background 3
- 2.1 5G NR . . . 3
  - 2.1.1 5G NR RAN Protocol Architecture . . . 4
  - 2.1.2 5G NR Control Plane . . . 6
  - 2.1.3 Radio Resource Control . . . 6
  - 2.1.4 Synchronization Signals in 5G NR . . . 7
- 2.2 Handovers in 5G NR . . . 8
  - 2.2.1 Handover performance metrics . . . 10
- 2.3 Machine Learning Models . . . 10
  - 2.3.1 Supervised Learning . . . 10
  - 2.3.2 Unsupervised Learning . . . 11
  - 2.3.3 Reinforcement Learning (RL) . . . 12
  - 2.3.4 Evaluation Metrics . . . 12
- 2.4 Related Work . . . 13
  - 2.4.1 Supervised Learning for Handover Prediction . . . 13
  - 2.4.2 Reinforcement Learning Approaches . . . 14
  - 2.4.3 ML for Specialized Scenarios . . . 14
  - 2.4.4 AI/ML for Beam Management in 5G-Advanced . . . 14
  - 2.4.5 Summary . . . 14

3 Methodology 15
- 3.1 DB File Generation Using Redhawk . . . 15
- 3.2 Data Collection . . . 17
- 3.3 Feature Engineering and Selection . . . 17
- 3.4 Labeling . . . 17
- 3.5 Model Training . . . 18
- 3.6 Evaluation Metrics . . . 18

4 Results **19**
4.1 Random Forest Results . . . 19
4.1.1 Confusion Matrices . . . 19
4.1.2 ROC Curves . . . 21
4.2 XGBoost Results . . . 22
4.2.1 Confusion Matrices . . . 22
4.2.2 ROC Curves . . . 23

5 Discussion **25**
5.1 Results . . . 25
5.1.1 Interpretation of Results From Random Forest . . . 25
5.1.2 Discussion on the Results Obtained From XGBoost . . . 26
5.2 Method . . . 28
5.2.1 Limitations in Dataset Separability . . . 28
5.2.2 Comparison of XGBoost Results Across Scenarios . . . 28
5.2.3 Comparison of Random Forest Results Across Scenarios . . . 29
5.2.4 Reliability, Replicability, and Validity . . . 29
5.3 The Work in a Wider Context . . . 29

6 Conclusion **32**
6.1 Summary of Aim and Research Questions . . . 32
6.2 Answers to Research Questions . . . 32
6.3 Implications . . . 33
6.4 Future Work . . . 34

Bibliography **35**

List of Figures

2.1 5G NR RAN architecture illustrating the split between Central Unit (CU) and Distributed Units (DU), and the main interfaces (NG, Xn, F1) [3] 4
2.2 5G NR protocol stack in user and control planes between UE, gNB, and AMF . . . 5
2.3 Legacy L3 Handover Signaling Diagram 8
2.4 NR measurement model 9

4.1 Confusion matrices for Random Forest (3 features) — Pedestrian scenarios. 20
4.2 Confusion matrices for Random Forest (3 features) — Vehicular scenarios. 20
4.3 ROC curves for Random Forest (3 features) — Pedestrian scenarios. 21
4.4 ROC curves for Random Forest (3 features) — Vehicular scenarios. 21
4.5 Confusion matrices for XGBoost (3 features) — Pedestrian scenarios. 22
4.6 Confusion matrices for XGBoost (3 features) — Vehicular scenarios. 23
4.7 ROC curves for XGBoost (3 features) — Pedestrian scenarios. 23
4.8 ROC curves for XGBoost (3 features) — Vehicular scenarios. 24

5.1 Random Forest confusion matrices in the 10UE pedestrian scenarios. 26
5.2 XGBoost confusion matrices in the 10UE vehicle scenarios. 28

List of Tables

3.1 Redhawk simulation parameters and system configuration 16

4.1 Random Forest hyperparameters tunning for both pedestrian and vehicular scenarios. 19
4.2 XGBoost hyperparameters used for pedestrian and vehicular scenarios. 22

1 Introduction

1.1 Motivation

Ensuring smooth connectivity as users move between base stations remains a vital challenge for mobile networks. The current handover mechanisms, though effective, often react to changes rather than anticipate them. This reactive nature introduces latency, unnecessary handovers, and inefficiencies that degrade the overall user experience—problems that become more pronounced in scenarios involving high mobility or fluctuating signal environments. As mobile usage patterns grow increasingly dynamic, and the number of connected devices continues to rise (particularly with the expansion of IoT networks), the demand for smarter, more adaptive mobility management strategies becomes even more critical. AI and ML offer promising solutions to this problem, with the ability to analyze complex datasets, predict user movement, and adjust network behavior proactively. Leveraging these technologies could lead to significant improvements in handover performance, network efficiency, and user satisfaction. This work is driven by the opportunity to harness AI/ML techniques to refine L3 handover processes, reduce network overhead, and create a more seamless communication experience for users in a constantly changing mobile landscape.

1.2 Aim

The objective of this thesis is to investigate how Artificial Intelligence (AI) and Machine Learning (ML) methods can enhance mobility management within cellular networks, with particular emphasis on improving traditional Layer 3 (L3) handover processes. Central to this work is the development and evaluation of supervised learning models designed to predict handover outcomes using mobility-related data, with the aim of making L3 handover decisions more accurate and efficient.

In addition, this research seeks to highlight the inherent limitations of conventional L3 handover strategies, explore the potential advantages offered by AI/ML-driven optimization, and generate insights that could shape future advancements in intelligent mobility management—both in academic study and in real-world network applications.

1.3 Research questions

1. In what ways can AI and ML algorithms improve the accuracy and efficiency of L3 handover decisions in LTE and 5G networks?
2. Which mobility-related features have the greatest impact on predicting the success of L3 handover events?
3. In what ways do supervised learning models, such as XGBoost and Random Forest, outperform traditional threshold-based handover methods in terms of overall performance?
4. What limitations currently affect AI/ML-based approaches to L3 mobility management, and what promising avenues could future research explore to further advance intelligent mobility optimization?

1.4 Delimitations

This thesis focuses exclusively on the application of AI and ML algorithms to Layer 3 (L3) mobility management in LTE and 5G networks. While Layer 1 and Layer 2 Triggered Mobility (LTM) is briefly discussed in the theoretical background, no experimental implementation or evaluation of LTM is performed in this work.

The experimental scope is limited to supervised learning methods, specifically Random Forest and XGBoost classifiers. Other AI/ML approaches, such as reinforcement learning and deep learning, are discussed in the theoretical background and related work for completeness but are not implemented or evaluated here. Furthermore, the evaluation is conducted using existing L3 mobility datasets; no live network data collection or deployment in operational networks is included.

2 Theoretical Background

2.1 5G NR

The Fifth Generation (5G) New Radio (NR) system marks a significant leap forward in mobile wireless technology, evolving directly from the groundwork laid by Long Term Evolution (LTE). While LTE was instrumental in delivering high-speed mobile broadband to users around the world, the explosion of connected devices, the rising demand for ultra-reliable, low-latency services, and the growing popularity of data-heavy applications quickly exposed LTE's shortcomings. In particular, it became clear that greater flexibility, improved scalability, and enhanced spectral efficiency were necessary to keep pace with these rapidly changing requirements [1].

Evolution from LTE:
To overcome the constraints of previous wireless technologies, 5G NR was developed to operate across a vastly expanded frequency spectrum, ranging from below 1 GHz up to 52.6 GHz. This standard also introduced several innovative features, such as scalable numerology, adaptable subcarrier spacing, and a beam-focused, energy-efficient architecture. These enhancements make it possible for 5G NR to deliver not only high-speed broadband but also to accommodate massive Machine Type Communications (mMTC) and ultra-reliable, low-latency communications (URLLC) [1]. Another fundamental design principle was forward compatibility, allowing the system to flexibly adapt as new use cases and technologies emerge. This adaptability enables 5G NR to perform effectively [2] in a variety of settings, from dense cityscapes to large-scale industrial environments.

Driving Forces Behind 5G:
The development from LTE to 5G NR has been shaped by several important trends and evolving requirements. The explosive growth in data usage, driven by streaming services, cloud computing, and social networks, has put immense pressure on network capacity. Simultaneously, the expanding Internet of Things (IoT) has created a need for scalable, efficient ways to connect billions of affordable, low-power devices. New use cases such as autonomous vehicles, smart manufacturing, and remote healthcare have introduced a demand for extremely reliable, low-latency communications—capabilities that LTE simply could not offer. Experience from commercial deployments further highlighted the necessity

of reducing both capital and operating costs, boosting energy efficiency, and making more effective use of spectrum resources [2]. 5G NR also addresses industry-specific needs, such as supporting uplink-heavy traffic (like video uploads and industrial monitoring) and delivering highly accurate positioning, especially in industrial IoT scenarios [2].

5G Use-Cases:
The International Telecommunication Union (ITU) and 3GPP have defined three major categories of 5G use-cases, all of which are explicitly supported by 5G NR:

- **Enhanced Mobile Broadband (eMBB):** High-throughput, low-latency applications, such as ultra-HD video, XR, and cloud-based gaming, that demand gigabit data rates and seamless connectivity.
- **Massive Machine-Type Communications (mMTC):** Support for large-scale IoT deployments, enabling connectivity for billions of devices with diverse requirements for power, bandwidth, and coverage.
- **Ultra-Reliable Low-Latency Communications (URLLC):** Mission-critical services such as industrial automation, remote surgery, and autonomous vehicles, where deterministic, low-latency, and high-reliability connections are paramount.

Recent studies also note that, as 5G advances, new scenarios such as uplink-centric broadband communication (UCBC), real-time broadband communication (RTBC), and harmonized communication and sensing (HCS) are emerging, expanding the landscape of 5G use-cases even further [2].

In conclusion, 5G NR marks a paradigm shift in wireless communications by evolving the LTE framework to support a vastly more diverse and demanding range of applications, deployments, and industries. Its flexible, future-proof architecture ensures that the network can continue to adapt as technological and commercial requirements evolve [1, 2].

2.1.1 5G NR RAN Protocol Architecture

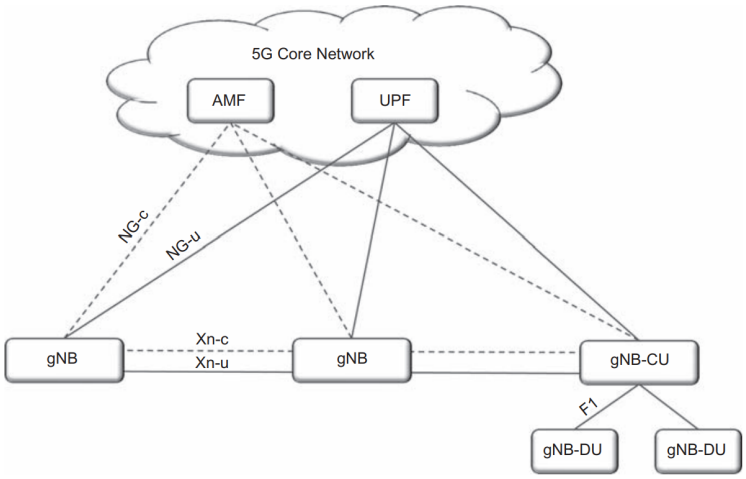


Figure 2.1: 5G NR RAN architecture illustrating the split between Central Unit (CU) and Distributed Units (DU), and the main interfaces (NG, Xn, F1) [3]

The 5G New Radio (NR) Radio Access Network (RAN) is designed to provide high flexibility, scalability, and service differentiation to support the diverse requirements of modern wireless networks. At the heart of the 5G NR RAN is the **gNB** (next-generation NodeB), which serves

NR user equipment by managing all radio-related functions for one or more cells [3]. The overall structure of the 5G NR RAN is illustrated in Figure 2.1, showing the split between the Central Unit (CU) and Distributed Unit (DU), along with the key interfaces that enable communication within the network and with the core.

A notable innovation in the 5G NR RAN is the logical split of the gNB into two components: a Central Unit (gNB-CU) and one or more Distributed Units (gNB-DUs). The **gNB-CU** is responsible for higher-layer protocols such as RRC, PDCP, and SDAP, and can be deployed centrally to manage multiple DUs. In contrast, the **gNB-DU** handles lower-layer protocols (RLC, MAC, PHY) and is typically positioned closer to the radio interface, enabling low-latency operations and flexible deployment strategies. This architecture supports advanced concepts such as centralized RAN (C-RAN), distributed RAN, or hybrid approaches, depending on service and network needs.

Communication within the RAN and to the core network is handled via standardized interfaces. The **NG interface** connects the gNB to the 5G Core Network, with the NG-c part for control plane (to the Access and Mobility management Function (AMF)) and NG-u for user plane (to the User Plane Function (UPF)). The **Xn interface** connects gNBs to each other, enabling seamless mobility, active-mode handover, and dual connectivity. The **F1 interface** supports the functional split between the gNB-CU and gNB-DU, allowing flexible placement and scaling of network functions.

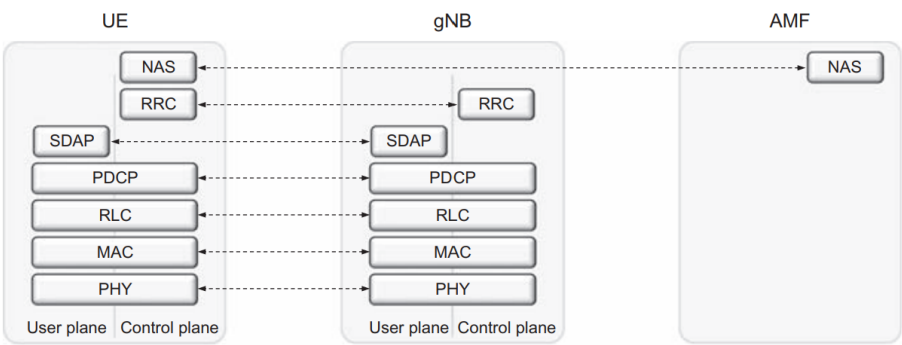


Figure 2.2: 5G NR protocol stack in user and control planes between UE, gNB, and AMF

The protocol stack of 5G NR RAN consists of several layers, as illustrated in Figure 2.2. The main protocol entities are:

- **Service Data Adaptation Protocol (SDAP)**
- **Packet Data Convergence Protocol (PDCP)**
- **Radio Link Control (RLC)**
- **Medium Access Control (MAC)**
- **Physical Layer (PHY)**

By introducing a logical separation between user plane and control plane, and enabling function splits at multiple levels, the 5G NR RAN architecture allows operators to deploy, manage, and optimize their networks for a wide variety of use cases, including enhanced mobile broadband, ultra-reliable low-latency communication, and massive machine-type communication. This modular and flexible architecture is a fundamental enabler for network slicing and the future evolution of mobile networks [3].

2.1.2 5G NR Control Plane

The 5G NR control plane consists of a set of dedicated protocols that manage all signaling and control communication between user equipment (UE), the radio access network (gNB), and the core network [3]. These protocols enable connection establishment, configuration, mobility management, security, and session maintenance, forming the backbone of intelligent and flexible network operation.

Key Control Plane Protocols:

- **Radio Resource Control (RRC):** RRC is the main protocol at the control plane for the interface between the UE and the gNB. It manages the setup, modification, and release of radio bearers, mobility procedures (including handover), security activation, measurement control, paging, and broadcast of system information. RRC also governs transitions between RRC states: RRC_IDLE, RRC_INACTIVE, and RRC_CONNECTED, determining how the UE interacts with the network and handles mobility.
- **Non-Access Stratum (NAS):** The NAS protocol runs between the UE and the core network (specifically, the AMF). It is responsible for functions such as authentication, security management, and registration. NAS messages are carried over the radio interface via RRC signaling.
- **Packet Data Convergence Protocol (PDCP):** PDCP handles header compression, encryption, integrity protection, and in-sequence delivery for control messages, ensuring secure and efficient transmission of signaling data.
- **Radio Link Control (RLC):** RLC segments, reassembles, and may retransmit control plane data, operating in various modes (Transparent Mode (TM), Unacknowledged Mode (UM), and Acknowledged Mode (AM)) to balance reliability and latency.
- **Medium Access Control (MAC):** MAC supports logical channel prioritization, scheduling, and multiplexing, ensuring timely and efficient delivery of control plane data alongside user plane traffic.
- **Physical Layer (PHY):** PHY encodes, modulates, and transmits the control information over the air interface, using dedicated control channels for signaling such as DCI (Downlink Control Information) and UCI (Uplink Control Information).

These protocols work together to guarantee reliable and secure signaling for connection management, mobility procedures, handover, and resource allocation. For example, an RRC connection setup or handover event may involve the exchange of NAS, RRC, and PDCP messages, all mapped onto the RLC, MAC, and PHY layers for transmission between the UE and gNB.

2.1.3 Radio Resource Control

The Radio Resource Control (RRC) protocol is a critical part of the 5G NR control plane. It manages how the user equipment (UE) and the radio access network communicate for tasks such as connection setup, mobility, paging, and security [4]. One of the biggest innovations in 5G is the way the RRC state machine is designed to support a much wider variety of use cases and device behaviors than previous generations.

Unlike LTE, which relied on just two primary RRC states (Idle and Connected), 5G NR introduces a third, highly flexible state called **RRC Connected Inactive**. The state machine now consists of:

- **RRC Idle:** The UE is not actively connected and saves power. It can be reached through paging, but does not store any connection context.

- **RRC Connected:** The UE has an active connection, and the context is stored both in the UE and in the network. All data transfer and mobility are managed in this state.
- **RRC Connected Inactive:** This new state maintains the connection context in both the UE and the network, and maintains the link to the core network. The UE can move within a tracking area and be reached by paging, but does not require a full connection setup to send or receive data, which drastically reduces signaling, power usage, and connection latency.

The introduction of RRC Connected Inactive is especially important for devices and applications that need to transmit small, infrequent packets (such as IoT sensors), as well as for users who move frequently but only require brief or occasional network access. This state is also highly configurable, allowing the network to tailor device behavior—such as how often the UE wakes up, how paging is handled, or how quickly it can transition to active communication—based on specific service requirements.

Transitions between these RRC states are optimized for speed and efficiency. When the UE in RRC Connected Inactive needs to become fully active, it can do so rapidly, since all necessary context is already available. Performance analyses show that using this state can significantly cut down signaling overhead, reduce control plane latency, and improve battery life compared to earlier approaches in LTE.

Overall, the flexibility and configurability of the 5G NR RRC state machine provide a strong foundation for supporting the broad spectrum of services and applications expected in future mobile networks [4].

2.1.4 Synchronization Signals in 5G NR

Synchronization signals are fundamental for initial access, cell search, and mobility procedures in 5G NR networks. They enable user equipment (UE) to detect, identify, and synchronize with a 5G cell before any data transmission occurs [5].

5G NR introduces a new synchronization signal block (SSB), which consists of the Primary Synchronization Signal (PSS), Secondary Synchronization Signal (SSS), and the Physical Broadcast Channel (PBCH). The PSS enables the UE to acquire symbol timing and determine the physical layer identity, while the SSS provides frame timing and further identification information. The PBCH conveys essential system information required for the UE to establish a connection with the network.

The SSBs are transmitted directionally in different beams, particularly important in higher frequency (mmWave) deployments, where beamforming is used to combat signal attenuation. This approach allows the UE not only to synchronize with the cell but also to determine the best reception beam during the initial access phase.

The synchronization procedure begins with the UE scanning for PSS and SSS patterns in the received signal. Once detected, the UE can determine the subcarrier spacing, symbol and frame timing, and cell identity. The UE then decodes the PBCH to receive system information such as bandwidth, subcarrier spacing, and scheduling information for further system information blocks.

Compared to LTE, 5G NR's design offers greater flexibility by supporting different numerologies and enabling synchronization across a wider range of frequencies. This makes initial access and handover more robust, even in challenging propagation environments and dense deployment scenarios.

In summary, synchronization signals in 5G NR play a critical role in enabling fast and reliable initial access, beam selection, and mobility, ensuring that UEs can efficiently and seamlessly connect to 5G networks [5].

2.2 Handovers in 5G NR

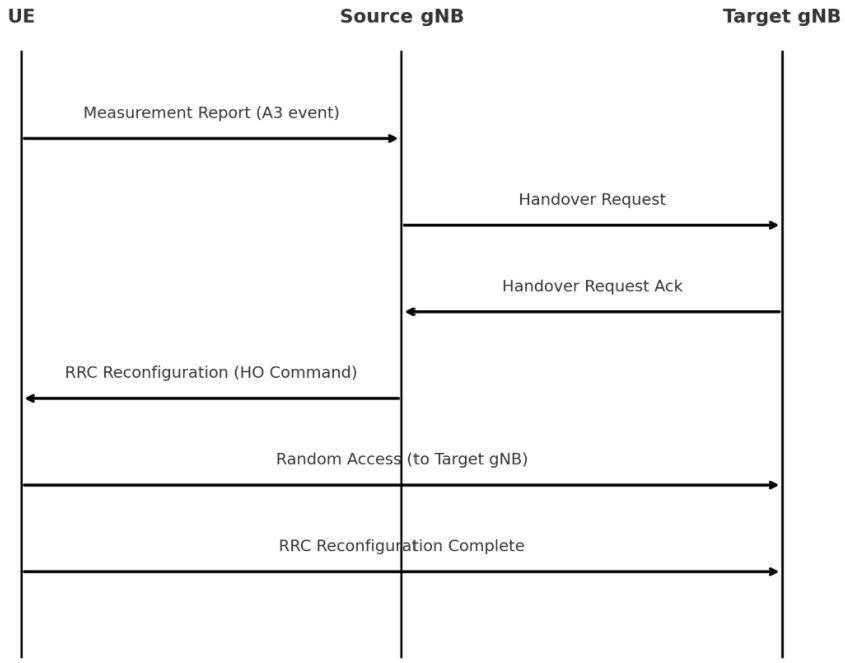


Figure 2.3: Legacy L3 Handover Signaling Diagram

In 5G New Radio (NR), legacy Layer 3 (L3) handovers typically depend on measurements of downlink signal quality, such as the Synchronization Signal Block (SSB) and Channel State Information Reference Signal (CSI-RS). User Equipment (UE) regularly monitors these signals from both the serving cell and neighboring cells to maintain a stable and high-quality connection. These measurements undergo filtering at both the physical (L1) and Radio Resource Control (L3) layers to ensure reliability despite fluctuations in the channel conditions [6].

The classical L3 handover process involves several key steps [6]:

- **Measurement and Filtering:** The UE constantly measures the quality of signals (SSB and CSI-RS) from surrounding cells. Measurements pass through L1 and L3 filters to smooth out short-term fluctuations.
- **Event Triggering (Event A3):** Event A3 is one of the standardized RRC measurement events defined by 3GPP. It is triggered when the received signal quality (RSRP or RSRQ) of a neighboring cell becomes better than that of the serving cell by a predefined offset. This makes A3 a central mechanism for mobility management and handover detection in LTE and NR. The handover process is triggered once this condition is met.
- **Time-to-Trigger (TTT):** To prevent frequent and unnecessary handovers ("ping-pong"), the UE waits for a predefined duration, called Time-to-Trigger, ensuring the improved signal condition is stable.
- **Measurement Report (MR):** Once the signal improvement persists through the TTT duration, the UE sends a Measurement Report to its current serving base station (gNB), informing it about the superior neighboring cell.
- **Handover Decision:** Upon receiving the MR, the serving gNB evaluates the request, ensuring the target cell has sufficient resources to accept the UE.

- **Handover Command (RRC Reconfiguration):** If the handover is approved by the target cell, the serving gNB sends an RRC Reconfiguration command to the UE, instructing it to switch to the target cell.
- **Execution and Completion:** The UE disconnects from the serving cell and establishes a connection with the target cell. The handover procedure completes successfully when the UE sends an RRC Reconfiguration Complete message to the target gNB. This well-established handover process ensures robust mobility management while optimizing network resources and maintaining seamless connectivity.

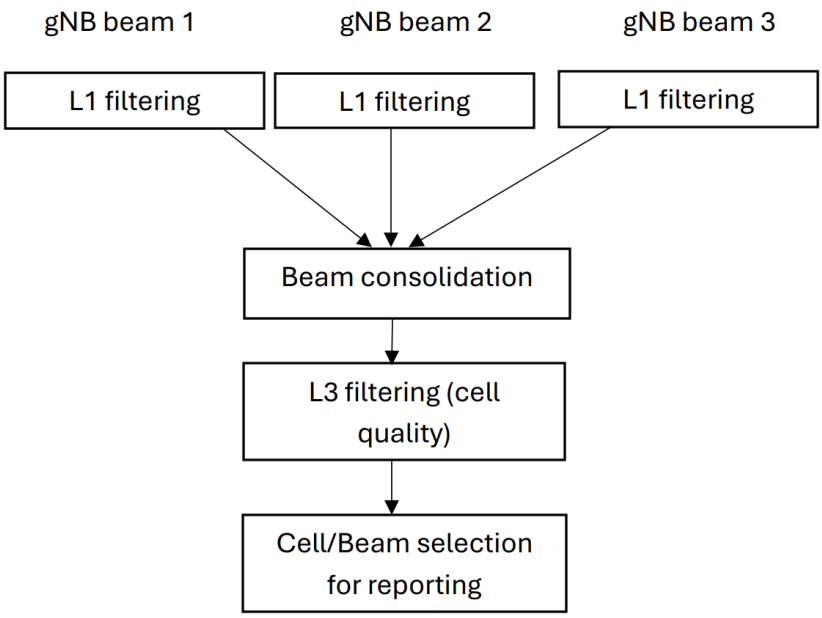


Figure 2.4: NR measurement model

Figure 2.4 illustrates the beam-based measurement and event evaluation procedure in 5G NR. Multiple beams transmitted by the gNB are first processed through Layer 1 (L1) filtering at the UE to mitigate fast fading and ensure signal stability. These filtered measurements are then consolidated and passed to Layer 3 (L3) filtering modules for further smoothing and evaluation. The top path represents cell-quality evaluation used for event triggering (e.g., A3), while the lower path supports beam-specific decisions such as beam selection for reporting. This modular structure enables accurate and timely mobility decisions under highly directional beamforming conditions [6]. The measurement and event evaluation procedure begins when the UE receives downlink signals from multiple gNB beams, including Synchronization Signal Blocks (SSB) and Channel State Information Reference Signals (CSI-RS). Each beam is individually processed through Layer 1 (L1) filtering to reduce fast fading effects. These per-beam measurements are then consolidated and passed through Layer 3 (L3) filtering to derive a stable estimate of overall cell quality. The filtered signal metrics are evaluated against the predefined A3 event condition, which is triggered when a neighboring cell's signal exceeds that of the serving cell by a specified offset. Upon triggering, the UE starts a Time-to-Trigger (TTT) timer to ensure signal stability and prevent unnecessary handovers. If the condition still holds after the timer expires, the UE transmits a Measurement Report (MR) to the serving gNB. The gNB then evaluates the report, considering network resources and mobility policy, to determine whether a handover should be initiated [6].

The Downlink and Uplink Decoupled (DAPS) handover procedure consists of three phases: preparation, execution, and completion [7]. In the preparation phase, the UE per-

forms measurements and sends a report to the serving gNB once conditions are met (e.g., Event A3). The serving gNB coordinates with the target gNB and sends a handover command to the UE. The key innovation appears in the execution phase, where the UE remains connected to the serving gNB while establishing a new connection with the target gNB. Dual protocol stacks are temporarily maintained to ensure continuous data flow. In the completion phase, data forwarding and path switching occur, and the UE eventually releases the old connection, completing the seamless handover process.

Conditional Handover (CHO), introduced in 3GPP Release 16, enhances reliability by decoupling handover preparation and execution [8]. The UE receives preconfigured handover commands from the serving gNB with conditions that must be met (e.g., RSRP, RSRQ thresholds). The UE continues communicating with the serving gNB until those conditions are fulfilled. Once met, it autonomously initiates the handover. CHO also includes fallback strategies to alternate candidates if the first attempt fails, minimizing handover failures and latency in highly dynamic environments.

2.2.1 Handover performance metrics

The performance of a handover (HO) procedure can be evaluated using various key metrics. Among these are the **handover failure (HOF) rate** which represents the proportion of failed handovers relative to the total number of attempted handovers which may occur due to the following radio link failures, synchronization errors or random access failures. The **handover frequency (HO rate)** indicates how often handovers occur per unit of time, with higher rates typically associated with smaller cells or higher user velocities. The **ping-pong (PP) rate** measures how frequently users rapidly switch back and forth between two cells within a short time frame, indicating unnecessary signaling overhead. The **energy consumption** quantifies the amount of energy used during the handover process which is critical for battery-powered devices. The **handover success rate** reflects the proportion of successful handovers compared to total attempts, indicating the robustness of the handover process. Additionally, **data latency** measures the delay in data transfer caused by the handover, while **handover interruption time** quantifies the duration during which the user experiences interruption in data connectivity due to the handover process [6].

2.3 Machine Learning Models

Machine learning (ML) encompasses various algorithms and statistical techniques that enable systems to automatically learn patterns from data and improve from experience without explicit programming. ML methods are broadly classified into supervised learning, unsupervised learning, and reinforcement learning, each suitable for distinct types of data and problems [9].

2.3.1 Supervised Learning

Supervised learning involves training models using labeled datasets, where each input sample has a corresponding target output. In this thesis, supervised learning is used to predict whether a handover will occur or not , based on features extracted from simulated mobility data. Each training sample includes input features such as the serving cell ID, best cell ID, and difference of RSRP values of serving cell and best cell. The output label is binary, indicating whether a handover occurred (1) or not (0). These labeled samples enable the models to learn patterns that distinguish between handover and non-handover cases. Once trained, the models can generalize to new data and support real-time handover decision-making. This approach offers a data-driven alternative to traditional threshold-based mobility management. Common supervised learning algorithms include the following:

2.3.1.1 Decision Trees (DT)

Decision trees partition the input space by recursively applying binary conditions on features, forming a hierarchical tree. Each internal node performs a test, branches represent outcomes, and leaves hold class labels. A common impurity measure is the Gini index:

$$Gini = 1 - \sum_{i=1}^{C} p_i^2$$

where p_i is the proportion of samples from class i, and C is the total number of classes. Decision trees are simple and interpretable but prone to overfitting [9]. They also form the foundation for ensemble methods such as Random Forest and XGBoost.

2.3.1.2 Random Forest (RF)

Random Forests are ensemble models that construct multiple decision trees using bootstrapped datasets and aggregate their predictions. Each tree uses a random subset of features at each split, which reduces correlation between trees. The final prediction is:

$$\hat{y} = \text{majority_vote}(T_1(x), T_2(x), \ldots, T_B(x))$$

This bagging approach improves accuracy, reduces variance, and avoids overfitting. Random Forest was one of the two main algorithms applied in this thesis, chosen for its robustness and high performance on structured mobility datasets [10].

2.3.1.3 Extreme Gradient Boosting (XGBoost)

XGBoost is an optimized gradient boosting algorithm that builds trees sequentially, with each tree correcting the errors of its predecessor. Its objective function combines loss and regularization:

$$\mathcal{L} = \sum_{i=1}^{n} L(y_i, \hat{y}_i^{(t)}) + \sum_{k=1}^{t} \Omega(f_k), \quad \Omega(f_k) = \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2$$

XGBoost uses shrinkage (learning rate), subsampling, and regularization to prevent overfitting and improve performance [11]. It was also used in this thesis as a core model due to its efficiency and strong predictive accuracy on handover datasets.

2.3.1.4 Support Vector Machines (SVM)

SVMs aim to find the optimal hyperplane that separates two classes with the largest margin. For linearly separable data, the optimization problem is:

$$\min_{\mathbf{w},b} \frac{1}{2}\|\mathbf{w}\|^2 \quad \text{subject to } y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geqslant 1$$

Kernel functions (e.g., RBF, polynomial) extend SVMs to non-linear classification tasks. SVMs are known to generalize well in high-dimensional spaces but can become computationally expensive for large datasets [12]. In this thesis, SVM is only included for background completeness; the experiments focus on Random Forest and XGBoost.

2.3.2 Unsupervised Learning

Unsupervised learning models work with unlabeled data, aiming to uncover patterns or hidden structures without predefined outputs. Although not applied in the experiments of this thesis, these methods are important for context in mobility research.

- **K-means Clustering**: It partition the data into k clusters by minimizing intra-cluster variance:

$$\min_{C} \sum_{i=1}^{k} \sum_{x \in C_i} \|x - \mu_i\|^2$$

  where μ_i is the centroid of cluster C_i. The K-means are computationally efficient but sensitive to initialization and the choice of k [9].

- **Principal Component Analysis (PCA)**: reduces the dimensionality of the data set by projecting the data onto principal components that capture the largest variance. This simplifies the dataset while retaining the most important information [13].

2.3.3 Reinforcement Learning (RL)

Reinforcement Learning (RL) is based on trial-and-error interaction with an environment. At each time step, an agent observes a state s, selects an action a, and receives a reward r. The objective is to maximize cumulative reward by learning a policy $\pi(a|s)$. A key concept is the action-value function:

$$Q(s, a) = r + \gamma \max_{a'} Q(s', a')$$

where $\gamma \in [0, 1]$ is a discount factor that balances the importance of immediate versus future rewards.

Common RL algorithms include:

- **Q-Learning**: a model-free method that updates Q-values iteratively using observed transitions.
- **Deep Q-Networks (DQN)**: extend Q-learning by using neural networks to approximate the Q-function in high-dimensional spaces.
- **Policy Gradient Methods**: optimize policies directly by computing gradients of expected return with respect to policy parameters.

RL has shown potential in mobility management, especially for adaptive handover optimization in dynamic environments [14]. However, due to its complexity and high resource requirements, RL was not used in this thesis and is included here only for background completeness.

2.3.4 Evaluation Metrics

Evaluation metrics are critical for assessing the performance and generalization ability of machine learning models. The choice of metric is particularly important when working with imbalanced datasets or when false positives and false negatives have different costs, as is the case in mobility prediction scenarios.

Accuracy measures the overall correctness of the model and is defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

While easy to interpret, accuracy is less informative for imbalanced datasets, where one class may dominate.

Precision is the proportion of true positive predictions among all positive predictions:

$$\text{Precision} = \frac{TP}{TP + FP}$$

High precision is important when false positives (e.g., predicting unnecessary handovers) are costly.

Recall, also known as sensitivity or true positive rate, is the proportion of actual positives correctly identified:

$$\text{Recall} = \frac{TP}{TP + FN}$$

Recall is critical when missing true events (e.g., failing to predict real handovers) must be avoided.

F1-score combines precision and recall using their harmonic mean:

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

It provides a balanced view, especially for imbalanced datasets.

F2-score places more weight on recall, which is suitable when missing an event is worse than a false alarm:

$$F_\beta = (1 + \beta^2) \cdot \frac{\text{Precision} \cdot \text{Recall}}{(\beta^2 \cdot \text{Precision}) + \text{Recall}} \quad \text{with } \beta = 2$$

Receiver Operating Characteristic (ROC) curve visualizes the trade-off between true positive rate and false positive rate. The Area Under the Curve (AUC) provides a single scalar to summarize performance across thresholds; a higher AUC indicates better model performance.

Mean Squared Error (MSE) is a common regression metric:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Though not used in this classification-focused thesis, MSE is important in general ML settings.

Conclusion: In this thesis, F1 score was used as the primary metric for handover (HO) prediction, and F2 score was used for unnecessary handover (UHO) prediction, reflecting the different cost trade-offs between precision and recall in those tasks.

2.4 Related Work

The application of machine learning (ML) techniques in handover management within LTE and 5G networks has been extensively explored in recent studies. Previous work can broadly be grouped into **supervised learning methods**, **reinforcement learning approaches**, and **AI-driven beam management for 5G-Advanced**.

2.4.1 Supervised Learning for Handover Prediction

Several studies have applied supervised ML algorithms to predict handovers and improve mobility robustness. In [15], a machine learning-based approach was proposed to enhance handover management in LTE networks with a focus on improving user quality of experience (QoE). Similarly, [16] evaluated multiple supervised learning algorithms to predict network coverage, showing that Random Forests offered high accuracy and robustness, making them suitable for facilitating handover decisions. Decision tree-based models have also been employed: for example, [17] optimized Random Access Channel (RACH) procedures for conditional handovers in 5G beamformed systems, demonstrating how tree-based prediction can improve mobility robustness. In another study, [18] applied ML to address signal obstructions during LTE handovers, showing that predictive models could mitigate service degradation caused by poor signal conditions.

2.4.2 Reinforcement Learning Approaches

Reinforcement learning (RL) has also been widely adopted for handover optimization. For instance, [19] applied asynchronous multiuser deep RL for handover control in wireless systems, showing that adaptive learning could improve throughput while reducing handover failures. Similarly, [20] provided an overview of RL algorithms for handover management in ultra-dense small-cell networks, highlighting their effectiveness in minimizing unnecessary handovers. Q-learning, a popular RL algorithm, was investigated in [21] to optimize mobility robustness in LTE Self-Organizing Networks (SON), reducing ping-pong effects and improving stability. These works underline RL's strength in environments where dynamic adaptation is crucial.

2.4.3 ML for Specialized Scenarios

More recent work has expanded the use of ML for specialized mobility challenges. In [22], ML-based handover decision models were tailored for non-terrestrial networks, showing benefits for satellite mobility. A broader survey of ML applications in 5G and beyond was conducted in [23], which outlined both the potential and limitations of applying ML, particularly emphasizing the promise of supervised and RL methods for handover optimization.

2.4.4 AI/ML for Beam Management in 5G-Advanced

With the emergence of 5G-Advanced, ML has also been integrated into beam management, which directly impacts mobility. The study in [24] examined ML-driven beam management from a standardization perspective, emphasizing improved handover efficiency and reduced latency. Similarly, [25] evaluated integrated ML models for beam management in 5G-Advanced and reported significant performance gains in handover reliability and user experience.

2.4.5 Summary

These studies provide compelling evidence of the potential of ML to improve handover performance by reducing failures, minimizing unnecessary handovers, and improving the user experience. However, most prior work has focused on either single techniques (decision trees, reinforcement learning) or *specific scenarios* (small cells, non-terrestrial networks, or beam management). Fewer studies have systematically compared **tree-based supervised methods such as Random Forest and XGBoost** under different mobility scenarios. This thesis addresses that gap by evaluating these models under both pedestrian and vehicular scenarios, for single-user (1UE) and multi-user (10UE) cases, thereby providing new insights into the strengths and limitations of supervised learning in mobility management.

3 Methodology

3.1 DB File Generation Using Redhawk

The simulation data used in this thesis was generated using Redhawk, a proprietary simulator developed by Ericsson to evaluate radio access network behavior under mobility conditions. Redhawk enables detailed and repeatable simulation of user equipment (UE) movement, handover procedures, and signal reporting within a 5G NR network. The simulation output was recorded in structured SQLite database files ('.db'), which served as the primary data source for training and evaluating machine learning models.

Each '.db' file corresponds to a specific simulation seed and scenario, and contains detailed logs of events such as measurement reports, beam selections, and handover decisions. Key parameters defined in each simulation include the network layout, user equipment (UE) type, motion model, carrier frequency, and handover configuration. The complete system model adopted in this work is summarized in Table 3.1, which provides an overview of the standardized settings applied across all generated datasets. Redhawk was selected for this study because it provides a controlled and repeatable simulation environment that closely reflects standardized 5G NR behavior. Unlike simplified mobility models, Redhawk captures beam-level measurements, cell-level consolidations, and the resulting handover decisions, ensuring that the generated data is suitable for both lower-layer and higher-layer mobility analysis.

In total, a set of simulation runs were conducted across pedestrian and vehicular scenarios with single-UE and multi-UE configurations. For each run, Redhawk outputs a '.db' file containing multiple tables, such as measurement logs, mobility events, and signaling messages. These tables were later joined and processed to create supervised learning datasets for model training. The database structure also ensures portability, enabling efficient parsing in Python through the 'SQLite3' interface. This was critical for preprocessing large volumes of simulation logs without performance bottlenecks.

A total of 30 different seeds were simulated, each with a runtime of 120 seconds. The network was deployed in an FR1 Urban Macro configuration consisting of 7 sites with 3 sectors per site, operating at a carrier frequency of 3.5 GHz with 30 kHz subcarrier spacing. The total bandwidth was 100 MHz, and the inter-site distance (ISD) was 500 meters, resulting in an effective cell radius of approximately 166.66 meters. A Time Division Duplex (TDD) scheme with a 3:2 ratio was applied for DL/UL transmission.

Table 3.1: Redhawk simulation parameters and system configuration

Parameters	**Values**
Simulation Files	
No. of Seeds	30
Simulation Time (s)	120
Network Deployment	
Deployment	FR1 Urban Macro (7 sites and 3 sectors per site)
Carrier Frequency	3.5 GHz with 30 KHz SCs
Bandwidth	100 MHz
Cell Radius	166.66 m (500 m ISD)
Duplex Scheme	TDD 3:2
gNB Antenna Height	25 m
No. of gNB Antennas	16
No. of UE Antennas	4
Noise factor (cell receiver)	5 dB
Noise factor (UE)	7 dB
No. of SSB Beams	4 SSB beams (20 ms SSB periodicity)
Max Downlink Transmission Power	20 Watt
UE Parameters	
No. of UEs	1 and 10
UE Speed	30 km/h (vehicle), 0.8333 m/s (pedestrian)
UE Motion	Random motion in straight directions
RRC Messages Exchanged	
Measurement Report	31 Octets
HO Command (RRC Reconfiguration)	891 Octets
HO Confirm (RRC Reconfiguration Complete)	2 Octets
Handover Event Parameters	
Event Name	A3 Event
Event Offset	2 dB
Event Time-to-Trigger	160 ms
Event Hysteresis	1 dB
L3 Filter Parameter (K)	4
Maximum Reported Cells	4

Key gNB parameters included:

- Antenna height: 25 meters
- Number of antennas: 16
- SSB configuration: 4 beams per gNB, with 20 ms periodicity
- Maximum downlink power: 20 Watts

The UEs were configured with 4 antennas and noise factor of 7 dB, while the gNB receiver had a noise factor of 5 dB. Both single-UE and multi-UE scenarios (1 UE and 10 UEs) were simulated, with pedestrian UEs moving at 0.8333 m/s and vehicular UEs at 30 km/h. UEs followed a random motion model in straight-line directions.

Handover triggering was based on Event A3 conditions with the following parameters:

- Event offset: 2 dB
- Time-to-trigger: 160 ms
- Hysteresis: 1 dB
- L3 filter parameter (K): 4
- Maximum number of reported cells: 4

In addition, Redhawk simulated standard RRC signaling procedures. The size of the signaling messages exchanged was:

- Measurement Report: 31 Octets
- HO Command (RRC Reconfiguration): 891 Octets
- HO Confirm (RRC Reconfiguration Complete): 2 Octets

All of these parameters are summarized in Table 3.1. The resulting '.db' files contained timestamped records of signal measurements (e.g., serving/best cell RSRP), event triggers, fulfilled neighbor lists, beam IDs, and mobility metadata, which were used for feature extraction, handover labeling, and machine learning model training.

This chapter outlines the methodology used to predict L3 handover events using machine learning in simulated cellular network scenarios. The workflow includes data collection, feature engineering, ground truth labeling, model training, and performance evaluation.

3.2 Data Collection

The simulation data was generated for four different scenarios (1 UE pedestrian, 10 UE pedestrian, 1 UE vehicle and 10 UE vehicle). For each scenario, measurement and event tables were extracted from the obtained .db files. Data preprocessing included column renaming and decoding binary fields.

3.3 Feature Engineering and Selection

A set of three features was considered in this study to evaluate the impact on handover prediction performance: servingCellId, bestCellId, rsrp_diff (i.e., the difference between RSRP values obtained for the serving cell and the best cell). Feature selection was motivated by the relevance of these parameters to the L3 handover decision in cellular networks. The set of features was systematically evaluated using the selected machine learning models and a comparative analysis of the performance of the model was performed for the four different scenarios.

3.4 Labeling

The handover events were identified by selecting column with `eventID == 30` in the event tables. The measurement and event tables were merged using time-based matching to align measurement samples with corresponding event occurrences. Based on this merging, a binary target label (`handover_occurred`) was assigned to each sample, where a value of 1 indicates detected handover event and 0 indicates no handover event.

3.5 Model Training

Two supervised machine learning models were used: Random Forest and XGBoost. Training and testing splits were performed per simulation seed, using 70% of the data for training and 30% for testing. Hyperparameters such as n_estimators and max_depth were tuned for each model. It should be noted that while the theoretical background covers supervised, unsupervised, and reinforcement learning approaches, the experimental scope of this thesis focuses exclusively on supervised methods—specifically Random Forest and XGBoost.

3.6 Evaluation Metrics

Performance was evaluated using the following metrics: accuracy, precision, recall, F1 score, ROC AUC, and confusion matrices.

The datasets, features, and preprocessing steps outlined in this chapter establish the experimental foundation. In the next chapter, these methods are applied to evaluate Random Forest and XGBoost across pedestrian and vehicular scenarios.

4 Results

This chapter presents the results obtained from the machine learning experiments using three input features: *servingCellId*, *bestCellId*, and *rsrp_diff*. The results are shown in terms of confusion matrices and ROC curves for both Random Forest and XGBoost classifiers. All results are presented factually, without interpretation, which is reserved for the discussion chapter.

4.1 Random Forest Results

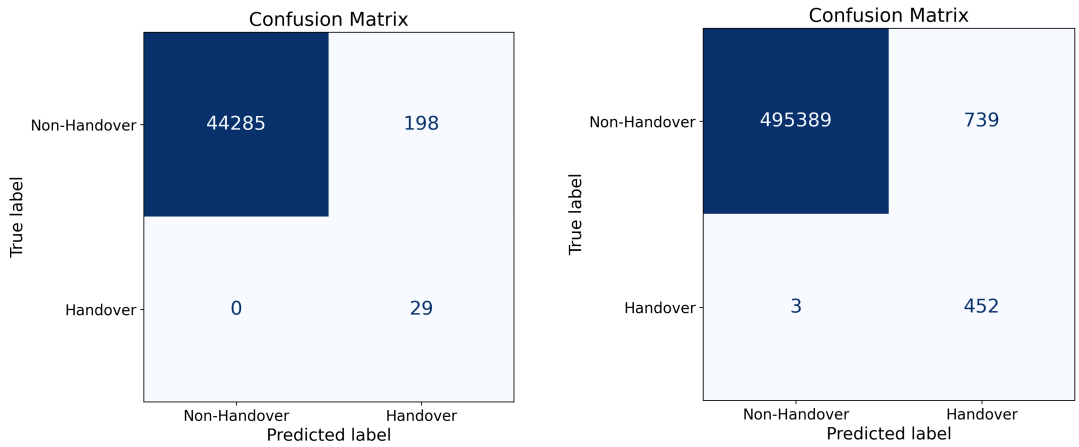
The Random Forest classifier was trained separately for pedestrian and vehicular scenarios. Different hyperparameter configurations were used to account for differences in data distribution and mobility characteristics. The chosen hyperparameters are summarized in Table 4.1.

Table 4.1: Random Forest hyperparameters tunning for both pedestrian and vehicular scenarios.

Scenario	**n_estimators**	**max_depth**
1UE Pedestrian	50	5
10UE Pedestrian	50	5
1UE Vehicle	10	5
10UE Vehicle	10	5

4.1.1 Confusion Matrices

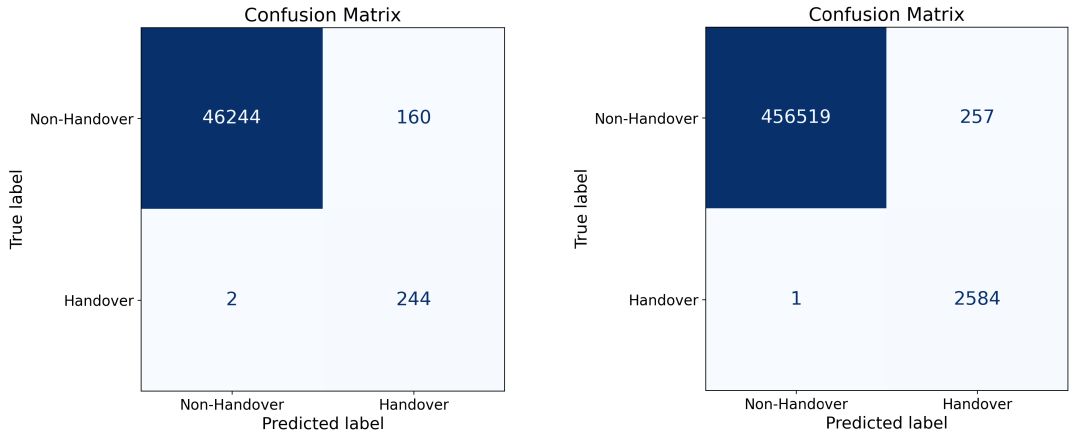
Figures 4.1 and 4.2 present the confusion matrices obtained with the Random Forest classifier for the four mobility scenarios: 1UE pedestrian, 10UE pedestrian, 1UE vehicle, and 10UE vehicle. These figures illustrate the classification outcomes in terms of true positives, true negatives, false positives, and false negatives for each mobility scenario.



(a) 1UE Pedestrian (b) 10UE Pedestrian

Figure 4.1: Confusion matrices for Random Forest (3 features) — Pedestrian scenarios.

The model detected all 29 handovers in the 1UE case (Recall = 100%) , with only 198 non-handovers misclassified as handovers, giving a very small false positive rate. In the 10UE case, it detected 452 out of 455 handovers (Recall 99%), but also produced 739 false positives, which can be explained by the increased interference and variability in a denser setup.



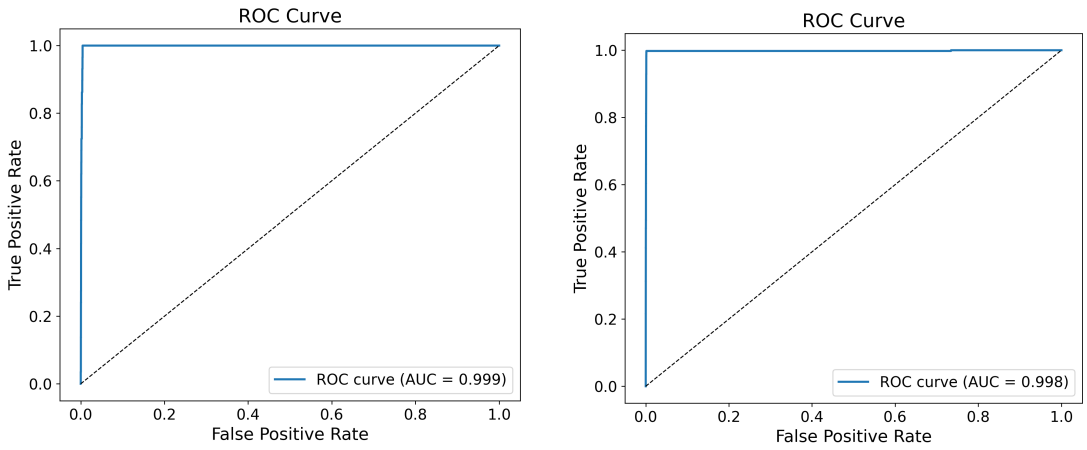
(a) 1UE Vehicle (b) 10UE Vehicle

Figure 4.2: Confusion matrices for Random Forest (3 features) — Vehicular scenarios.

In the 1UE case, the model detected 244 out of 246 handovers (99% recall) but also gave 160 false positives. This makes sense because at higher speeds the signal changes quickly, and the model sometimes mistakenly identifies those fluctuations as a handover. In the 10UE case, the results were even stronger: the model caught 2,584 out of 2,585 handovers (99% recall) and only produced 257 false positives out of more than 456,000 non-handovers. With multiple UEs moving simultaneously, the dataset contains more measurement samples and the mobility patterns become easier to distinguish from random fluctuations. This reduces noise and it helps the model to make more consistent and reliable predictions compared to single UE.

4.1.2 ROC Curves

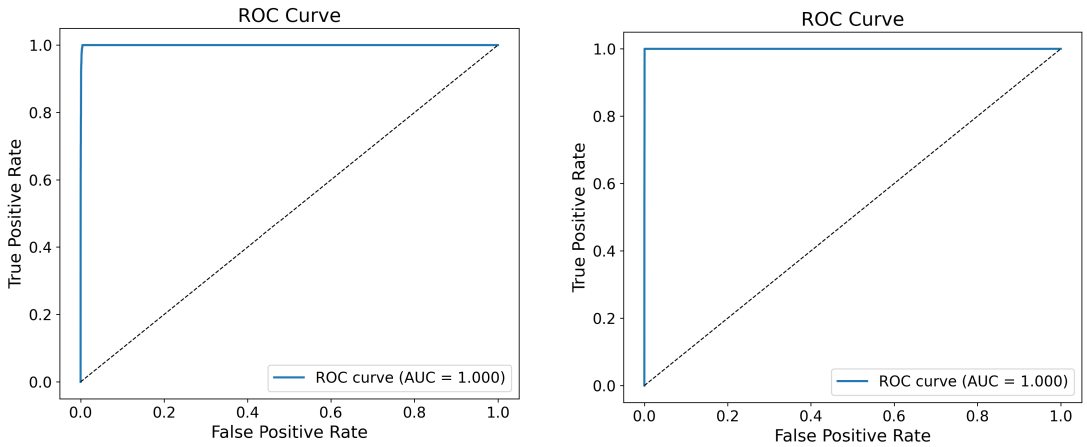
Figures 4.3 and 4.4 show the ROC curves of the Random Forest classifier for the four mobility scenarios: 1UE pedestrian, 10UE pedestrian, 1UE vehicle, and 10UE vehicle, respectively. The ROC curves give an overall picture of how well the classifier can separate handover from non-handover cases, with the AUC values provide a single measure of performance.



(a) 1UE Pedestrian (b) 10UE Pedestrian

Figure 4.3: ROC curves for Random Forest (3 features) — Pedestrian scenarios.

For the pedestrian cases, the ROC curves show almost perfect separation between handover and non-handover cases, with AUC values of 0.999 (1UE) and 0.998 (10UE). This means the model is highly reliable at distinguishing the two classes, even when more UEs are present.



(a) 1UE Vehicle (b) 10UE Vehicle

Figure 4.4: ROC curves for Random Forest (3 features) — Vehicular scenarios.

In the vehicular cases, the model reached an AUC of 1.000 for both 1UE and 10UE, indicating nearly perfect classification performance. Despite the higher variability caused by mobility, the Random Forest model was still able to almost flawlessly separate handovers from non-handovers.

4.2 XGBoost Results

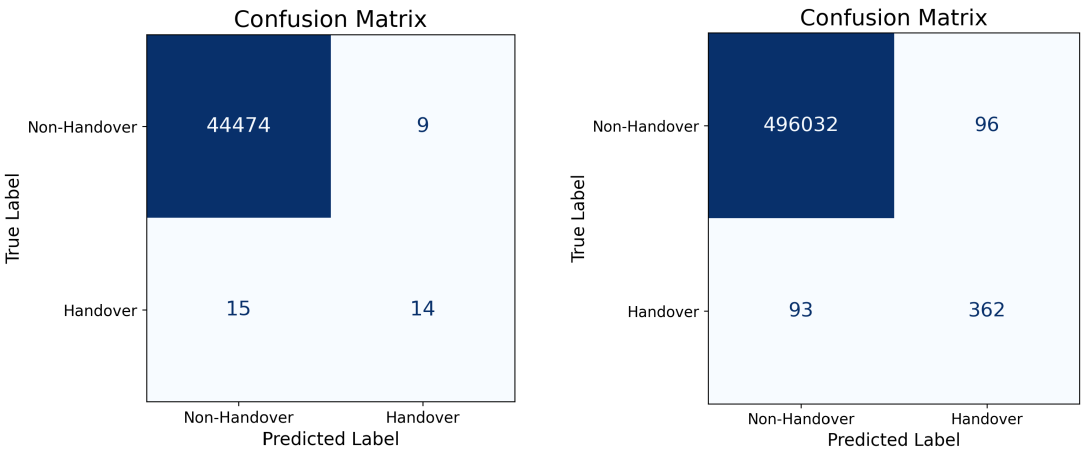
The XGBoost classifier was also trained separately for pedestrian and vehicular scenarios. As with the Random Forest model, different hyperparameter tunings were selected to account for the differences in mobility and data distribution. The chosen hyperparameters are summarized in Table 4.2.

Table 4.2: XGBoost hyperparameters used for pedestrian and vehicular scenarios.

Scenario	n_estimators	max_depth	learning_rate
1UE Pedestrian	100	5	0.5
10UE Pedestrian	100	5	0.5
1UE Vehicle	10	2	0.05
10UE Vehicle	10	2	0.05

4.2.1 Confusion Matrices

Figures 4.5 and 4.6 show the confusion matrices obtained with the XGBoost classifier for the four scenarios.

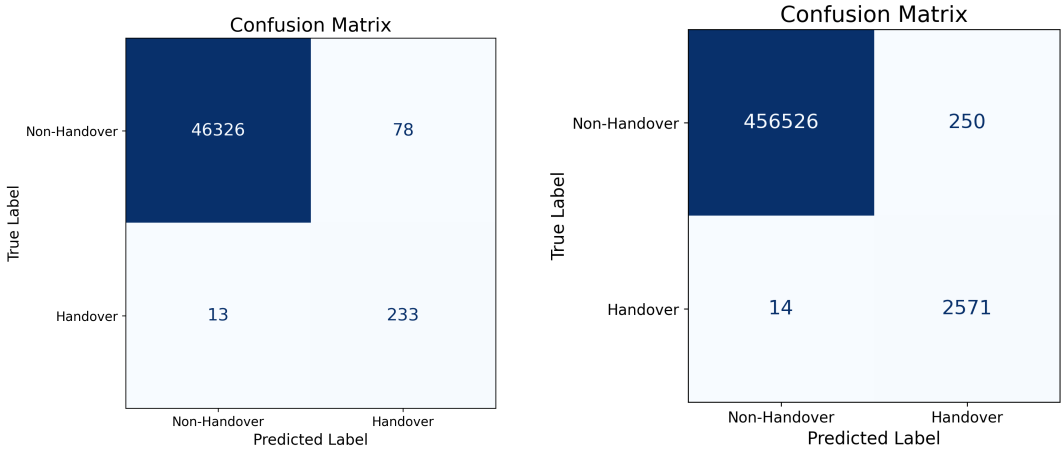


(a) 1UE Pedestrian (b) 10UE Pedestrian

Figure 4.5: Confusion matrices for XGBoost (3 features) — Pedestrian scenarios.

In the 1UE pedestrian case, the model correctly identified 14 out of 29 handovers (Recall 48%) while producing 15 false negatives. It made only 9 false positives out of more than 44,000 non-handovers, giving a precision of 61% and an F1-score of 0.54. This shows the model is conservative, predicting handovers only when it is confident, but recall remains modest.

In the 10UE pedestrian case, the model detected 362 out of 455 handovers (Recall 80%) while missing 93, and made only 96 false positives among nearly 500,000 non-handovers. This corresponds to a precision of 79% and an F1-score of 0.79. Compared to 1UE, recall and F1 are much higher, while precision remains strong. The denser environment in 10UE provides clearer signal variation patterns for the model to learn, enabling it to detect more handovers with relatively few false alarms.



(a) 1UE Vehicle (b) 10UE Vehicle

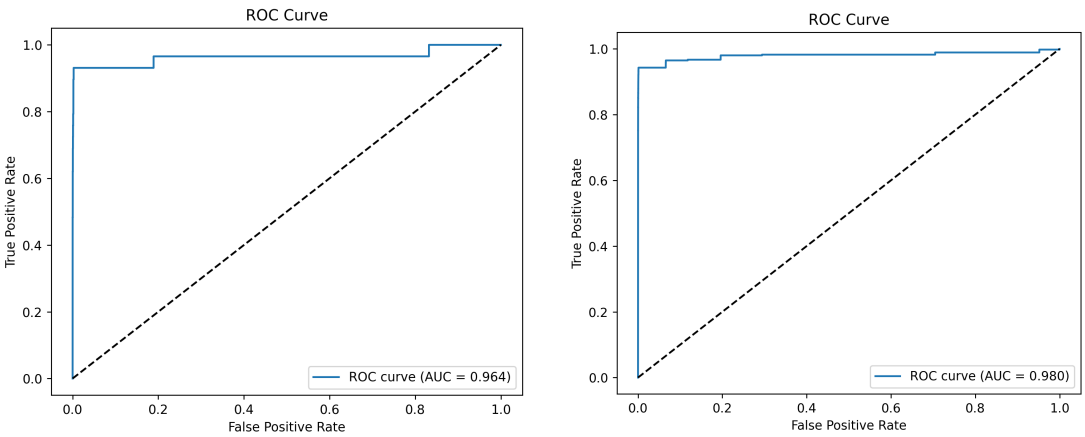
Figure 4.6: Confusion matrices for XGBoost (3 features) — Vehicular scenarios.

In the 1UE vehicle case, the model correctly detected 233 out of 246 handovers (Recall 95%) while missing 13, and produced 78 false positives out of more than 46,000 non-handovers. This corresponds to a precision of 75% and an F1-score of 0.84. These results show that the model is effective in capturing most handovers, though the fast channel variations at vehicular speed still lead to some false positives and missed handovers.

In the 10UE vehicle case, the performance improves further: the model correctly identified 2,571 out of 2,585 handovers (Recall 99.5%) while missing only 14, and made just 250 false positives among more than 456,000 non-handovers. This yields a precision of 91% and an F1-score of 0.95. The higher number of UEs provides more consistent handover patterns, enabling the model to achieve both very high recall and strong precision.

4.2.2 ROC Curves

Figures 4.7 and 4.8 show the ROC curves obtained with the XGBoost classifier for the four scenarios.

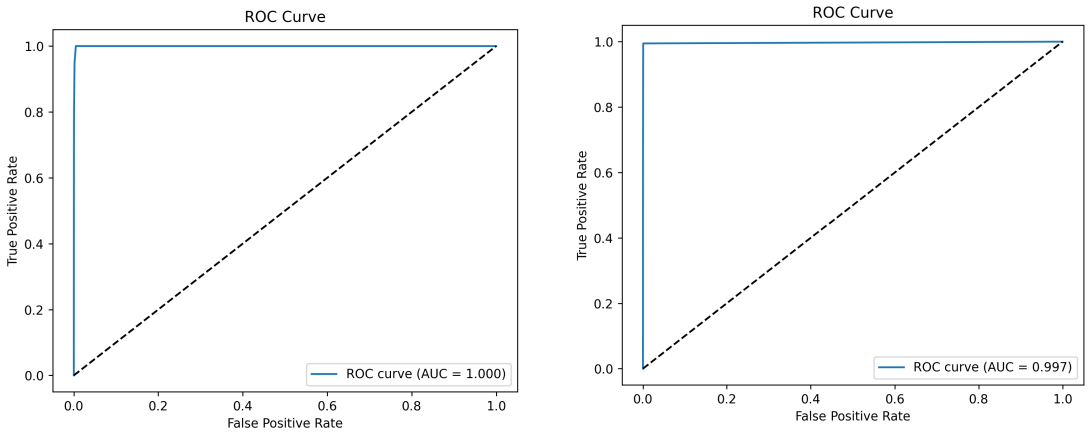


(a) 1UE Pedestrian (b) 10UE Pedestrian

Figure 4.7: ROC curves for XGBoost (3 features) — Pedestrian scenarios.

For the 1UE pedestrian case, the ROC curve shows an AUC of 0.964, which indicates that the model can reliably distinguish between handover and non-handover events, though not

perfectly. In the 10UE pedestrian case, the AUC increases to 0.980, showing even stronger separation between the two classes. The improvement reflects the richer and more consistent signal variation patterns available when multiple UEs are present, which helps the model learn clearer decision boundaries. Overall, both cases demonstrate excellent classification performance, with the 10UE scenario achieving the highest discrimination ability.



(a) 1UE Vehicle (b) 10UE Vehicle

Figure 4.8: ROC curves for XGBoost (3 features) — Vehicular scenarios.

In the 1UE vehicle case, the model achieved an AUC of 1.000, showing that it can perfectly separate handover from non-handover events despite the rapid signal changes caused by mobility. In the 10UE vehicle case, the AUC is slightly lower at 0.997, but still extremely high, indicating excellent classification performance. The small drop is expected due to the added complexity of handling multiple UEs, which introduces more interference and variability. Nevertheless, the model maintains very strong discrimination capability, almost perfectly distinguishing between the two classes.

These results provide a quantitative picture of model behavior across scenarios. However, metrics alone do not reveal why certain values are high or low, or what they mean in practice. The next chapter therefore interprets these findings in light of dataset properties, theoretical principles, and mobility management requirements.

5 Discussion

5.1 Results

The evaluation of Random Forest and XGBoost produced strong results overall, but with clear differences between pedestrian and vehicular scenarios. Both models reached very high accuracy values, yet accuracy alone was often misleading, particularly due to the highly imbalanced datasets. For this reason, it is necessary to interpret accuracy alongside precision, recall, and F1-scores. This section discusses why these values were high or low in each scenario and what this implies for model behavior in mobility management.

5.1.1 Interpretation of Results From Random Forest

As reported in Chapter 4, the final chosen hyperparameters for Random Forest were the number of estimators and maximum depth mentioned in (Table 4.1), with max_depth fixed at 5. Other parameters were kept at default values.

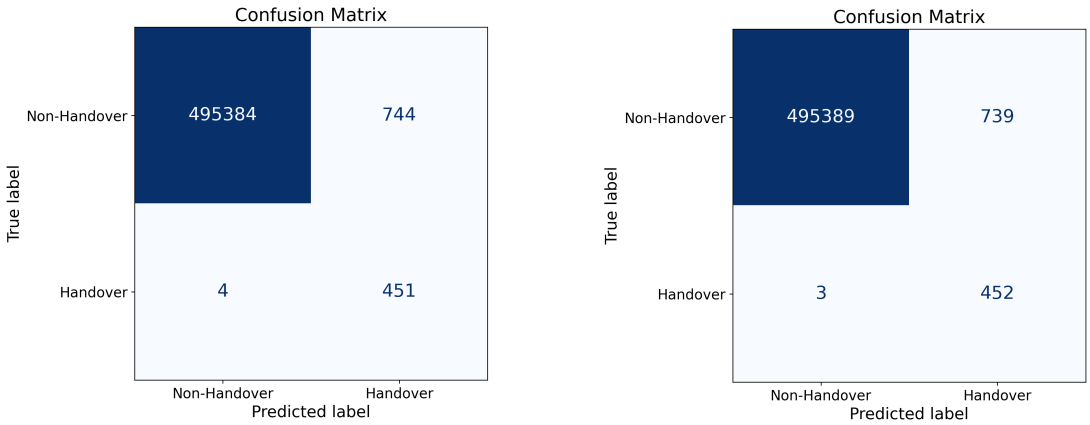
5.1.1.1 1UE pedestrian case:

With max_depth fixed at 5, accuracy stayed very high (0.99), but this was mainly due to class imbalance. Since real handovers were rare, the model could predict "handover" for almost everything and still achieve high accuracy. This explains why recall was perfect (1.0) — every handover was detected — but precision collapsed to 0.08–0.09, because almost all predicted handovers were false alarms. The F1-score was therefore very low ($\approx$0.15). Adding more trees did not change this, since the underlying problem was not lack of model complexity but the imbalance in the dataset. In practice, this would mean a network that avoids missed handovers but wastes resources by triggering many unnecessary ones.

5.1.1.2 10UE pedestrian case:

With max_depth = 5, the Random Forest model reached almost perfect accuracy overall. It detected almost all handovers (recall 0.99), but at the cost of many false positives, which kept precision low (0.38) and the F1 score moderate (0.55). The confusion matrices (Figures 5.1) make this clear: the model rarely misses a handover, but it tends to predict more handovers than actually occur. When the depth was set lower, such as 2 or 3, the model struggled to

capture the complexity of the data, and the classification of handover events was not satisfactory. However, increasing the depth further will lead to overfitting. The model performed very well on training data, but did not generalize as well on testing data. This makes max_depth = 5 a good middle ground, providing a balance between capturing enough detail and avoiding overfitting.



(a) 10UE pedestrian ($n_estimators = 5$, $max_depth = 5$).

(b) 10UE pedestrian ($n_estimators = 50$, $max_depth = 5$).

Figure 5.1: Random Forest confusion matrices in the 10UE pedestrian scenarios.

5.1.1.3 1UE vehicle case:

Here the situation improved. With max_depth = 5, recall was still perfect (1.0), but precision rose from 0.53 to 0.54, and the F1-score reached about 0.70. The reason is that in vehicular scenarios, RSRP values change more abruptly due to higher mobility. This creates clearer separations between handover and non-handover events, reducing false positives compared to the pedestrian case. Thus, the model benefited not from tuning but from the underlying signal characteristics of vehicular mobility.

5.1.1.4 10UE vehicle case:

This was the best-performing scenario. With max_depth = 5, Random Forest achieved almost perfect classification: accuracy = 1.0, recall = 1.0, precision = 0.91, and F1 = 0.95. In this case, accuracy was meaningful, since the model was highly correct across both classes. The improvement comes from two factors: (1) more UEs increased the dataset size, making it richer and less imbalanced, and (2) vehicular mobility made signal differences sharper and easier to separate. Together, these conditions allowed the Random Forest to classify handovers almost flawlessly.

5.1.2 Discussion on the Results Obtained From XGBoost

As reported in Chapter 4, the final chosen hyperparameters for XGBoost included the number of estimators, maximum depth, and learning rate (Table 4.2). All other parameters were kept at defaults. Unlike Random Forest, XGBoost's additional hyperparameter (learning rate) made it more sensitive to tuning. In each experiment, one parameter was varied while the others were fixed.

5.1.2.1 1UE pedestrian case:

When **learning rate was fixed at 0.1** and **estimators varied**, accuracy remained 1.0, but this was meaningless: the model exploited imbalance by predicting almost no handovers. Re-

call = 0.0, precision = 0.0, and F1 = 0.0. When depth was varied at fixed learning rate, recall rose slightly (0.0 → 0.31) and F1-score improved (0.0 → 0.41), but precision stayed 0.0. When learning rate itself was varied (0.1, 0.05, 0.01), results did not improve. In all cases, the model avoided false positives completely but missed almost every true handover. Thus, accuracy looked high but was deceptive, while recall, precision, and F1 showed complete failure in sparse, imbalanced data.

5.1.2.2 10UE pedestrian case:

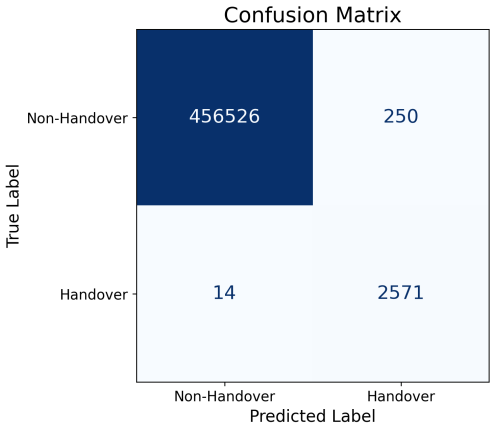
When **learning rate was fixed at 0.1** and **estimators varied**, accuracy was 1.0, this time reflecting more meaningful predictions. Recall improved to 0.60, precision was high (0.79), and F1 0.69. When depth was varied at fixed learning rate, recall increased further (0.60 → 0.73), F1 (0.69 → 0.77), while precision remained stable at 0.79. When learning rate was varied, recall improved slightly with lower values (0.66–0.73 at 0.05–0.01 learning rate) while precision still held steady at 0.79. In this richer dataset, accuracy was trustworthy, recall and precision both moderately high, and F1 indicated good balance. The reason for improvement was that more UEs reduced imbalance and noise, while lower learning rates refined decision boundaries.

5.1.2.3 1UE vehicle case:

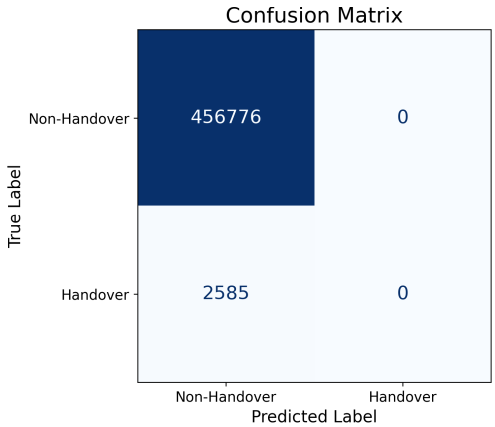
When **learning rate was fixed at 0.1** and **estimators varied**, accuracy was around 100%, recall rose (0.59 → 0.83), F1 improved (0.72 → 0.85), while precision decreased slightly (0.92 → 0.89). When depth was varied at fixed learning rate, recall jumped (0.59 → 0.95) and F1 rose (0.72 → 0.84), but precision dropped more (0.92 → 0.75). When learning rate was varied, results diverged sharply: at 0.1, recall was limited and precision stayed high; at 0.05, recall improved to 0.92–0.95 with F1-score of 0.84 and precision still acceptable; at 0.01 learning rate, recall stagnated and precision collapsed due to convergence issues. Thus, accuracy alone was always 100% but needed context: recall varied with tuning, precision dropped when recall improved, and F1 balanced the trade-off. The key factor was learning rate, which controlled stability.

5.1.2.4 10UE vehicle case:

When the **learning rate was fixed at 0.1** and **estimators varied**, accuracy was 100% and here it was fully justified: the predictions matched the reality almost perfectly. Recall reached 0.99, precision 0.91, and F1 0.95. However, reducing the learning rate revealed strong sensitivity. With a very small learning rate (0.01) and more estimators, the model collapsed completely, predicting no handovers at all. In this case, accuracy still appeared high (0.99) because of class imbalance, but recall and precision were both 0. By contrast, a moderate learning rate (0.05) with fewer estimators produced balanced results, with recall 0.99, precision 0.91, and F1 0.95. The confusion matrices in Figure 5.2 highlight this contrast: panel (a) shows the failed tuning run, while panel (b) shows the final configuration achieving near-perfect classification.



(b) XGB, 10UE vehicle ($n_estimators = 10$, $max_depth = 2$, $learning_rate = 0.05$).



(a) XGB, 10UE vehicle ($n_estimators = 20$, $max_depth = 2$, $learning_rate = 0.01$).

Figure 5.2: XGBoost confusion matrices in the 10UE vehicle scenarios.

5.2 Method

5.2.1 Limitations in Dataset Separability

A central methodological limitation of this study lies in the strong separability of the 1UE datasets. In particular, the 1UE vehicle scenario could be separated almost perfectly by applying a simple threshold to the *RSRP difference*. This meant that even simple models, and at times even heuristic rules, could achieve very high accuracy. As a result, both Random Forest and XGBoost frequently reached accuracy values of 1.0 in these cases. While this verified that the pipeline was correctly implemented and capable of learning from the data, it also reduced the scientific validity of the findings. When separability is trivial, the models are not required to capture deeper mobility patterns or make meaningful trade-offs, and the evaluation becomes less informative.

The pedestrian 1UE datasets revealed a complementary problem. Here, separability was not trivial, but class imbalance dominated. Random Forest predicted nearly all events as handovers (high recall but low precision), and XGBoost predicted almost none (high apparent accuracy but recall = 0). In both cases, accuracy looked high but did not represent true performance. This highlights a broader point: accuracy can be misleading in imbalanced datasets, and more nuanced metrics like precision, recall, and F1 are required to correctly interpret results.

By contrast, the 10UE datasets were less easily separable and provided a more realistic benchmark. These scenarios exposed the trade-offs between precision and recall much more clearly and highlighted how parameter tuning, particularly for XGBoost's learning rate, affected performance. This suggests that the 10UE datasets, although still simulated, more closely represent real-world challenges. For future work, experiments should include additional scenarios or real-world measurement data to avoid trivial separability and ensure higher external validity.

5.2.2 Comparison of XGBoost Results Across Scenarios

XGBoost's performance varied strongly with scenario complexity. In 1UE vehicle, for example, configurations with $n_estimators = 20$, max_depth = 2, and learning rate = 0.1 achieved accuracy of 1.0, precision = 0.89, and recall = 0.95. This suggests that the dataset was nearly separable, with the model capturing almost all handovers while keeping false positives low.

In the 1UE pedestrian case, however, accuracy was again high but precision dropped to near zero with recall = 0, because the model defaulted to conservative predictions. Here, high accuracy was masking complete failure on the minority class.

In the 10UE pedestrian scenario, accuracy was still 1.0, but class metrics were more balanced: precision ≈0.79, recall 0.60, F1 0.69. This shows that the model captured most handovers but introduced false positives, which is realistic in mobility prediction. In the 10UE vehicle case, accuracy = 1.0, precision 0.91, recall 0.99, F1 0.95. These strong results show that separability dominated, but with a dataset size large enough to avoid imbalance. The comparison across scenarios therefore highlights that XGBoost's outcomes are largely determined by dataset richness and separability.

5.2.3 Comparison of Random Forest Results Across Scenarios

Random Forest showed similar patterns, but its recall-oriented bias made the effects clearer. In the 1UE vehicle case, with $n_estimators = 5$ and max_depth $= 3$, accuracy was 100%, precision = 0.91, recall = 1.0, F1 = 0.95. This high performance confirms that the dataset was trivially separable. In the 1UE pedestrian case, however, accuracy close to 99% masked very poor class-specific performance: precision = 0.08–0.09, recall = 1.0, F1 ≈0.15. Here, the model predicted almost everything as a handover. This again shows why accuracy alone is not a sufficient indicator.

In the 10UE pedestrian case, results improved with accuracy = 1.0, precision ≈0.36, recall ≈0.99, and F1 ≈0.53. This improvement reflected greater dataset richness, which allowed the model to learn more balanced decision boundaries. In the 10UE vehicle case, results matched those of XGBoost, with precision ≈0.91, recall = 1.0, and F1 ≈0.95. These findings confirm that Random Forest is effective in highly separable or richer datasets, but breaks down in sparse and imbalanced ones.

5.2.4 Reliability, Replicability, and Validity

From a replicability perspective, the experiments were reliable. The datasets and preprocessing pipeline were clearly defined, and repeated runs with different random seeds produced consistent outcomes. This indicates that the reported results can be reproduced by other researchers using the same simulation setup. Reliability was further supported by the use of multiple simulation seeds, which reduced random variation and confirmed stable patterns. Validity, however, is more problematic. In 1UE datasets, trivial separability and imbalance inflated accuracy and F1, making the results less credible. These results appear artificially strong and cannot be generalized to real-world conditions. The 10UE datasets, although still simulations, provided a more valid testbed by exposing meaningful trade-offs and reducing imbalance effects. External validity remains limited, since real mobility traces are more complex and include additional factors such as interference, user behavior, and heterogeneous cell deployments. Future research should therefore validate findings on real network measurement data to ensure that conclusions hold beyond the controlled simulation environment.

5.3 The Work in a Wider Context

The findings of this study have broader implications that extend beyond the technical evaluation of Random Forest and XGBoost. While the experiments were conducted in a controlled simulation environment, the outcomes touch on issues of practical deployment, ethical responsibility, and societal relevance.

Technical Implications

From a technical perspective, the results highlight both the potential and the limitations of using machine learning for mobility management. Random Forest demonstrated robustness and reliability, guaranteeing that no handovers were missed. This is an important feature for networks where reliability and continuity are paramount, such as for voice calls, real-time video, or critical IoT services. However, the model was inefficient in sparse datasets, producing large numbers of false positives that would translate into unnecessary handovers in a live network. Excessive handovers increase signalling overhead, reduce spectral efficiency, and can cause temporary degradation in user experience. XGBoost, in contrast, was more conservative and required careful tuning, particularly of the learning rate, to achieve balanced outcomes. With the right configuration, however, it achieved better precision–recall trade-offs, reducing false positives without missing too many true handovers. This suggests that different models may be suited for different deployment contexts: Random Forest in scenarios where reliability is non-negotiable, and XGBoost in denser urban networks where efficiency is equally important.

The results also underline the importance of dataset properties in determining performance. The 1UE scenarios, being artificially separable or dominated by imbalance, produced results that looked strong in terms of accuracy but were not meaningful in practice. By contrast, the 10UE scenarios highlighted realistic trade-offs between precision and recall, and more closely resembled the conditions that real networks face. This shows that the design of training datasets is as critical as the choice of algorithm, and that testing models only on trivial datasets risks overestimating their real-world utility.

Ethical Considerations

Beyond technical performance, the use of AI/ML in mobility management raises important ethical considerations. One major issue is transparency. Ensemble models such as Random Forest and XGBoost are often described as "black-box" methods, making it difficult for operators and engineers to understand why a particular handover was predicted. This lack of interpretability complicates troubleshooting, accountability, and trust. If a model incorrectly triggers handovers or fails to trigger them when required, it is crucial for network operators to be able to explain why. Research into interpretability techniques, such as SHAP or permutation importance, could therefore play an important role in making these models more transparent.

Fairness is another ethical dimension. Although this thesis did not explicitly investigate fairness across users, the issue is highly relevant. Models trained on data that over-represents certain mobility patterns may generalize poorly to others, potentially resulting in unequal quality of service. For instance, a model tuned to vehicular scenarios may work very well for cars but less well for pedestrians, creating disparities in user experience. Future work should therefore consider fairness explicitly, ensuring that predictive performance is consistent across different types of users and environments. Accountability is also an important ethical question. If machine learning models are integrated into live networks, responsibility for errors must be clearly defined. Missed handovers could lead to dropped calls or degraded service, while false positives could increase resource waste. In either case, network operators must ensure that adequate safeguards, fallback mechanisms, and monitoring systems are in place before deploying AI/ML for mobility management at scale.

Societal and Industrial Relevance

From a societal perspective, reliable and efficient mobility management is directly tied to user experience. The ability to maintain seamless connectivity is fundamental for modern communication services, ranging from video streaming and online gaming to emergency calls and connected vehicle applications. A failure to predict and execute handovers accurately can

have direct consequences for users, including frustration, service degradation, or even safety risks in critical scenarios such as vehicular communications. Conversely, models that trigger too many unnecessary handovers can reduce overall network efficiency, indirectly lowering quality of service for all users. For the telecom industry, the findings highlight both opportunity and caution. On the one hand, the study demonstrates that machine learning methods are capable of delivering very strong performance under realistic conditions, confirming that AI/ML can play a role in next-generation mobility management. On the other hand, the results also show that success is highly dependent on dataset design and careful hyperparameter tuning. This suggests that industrial adoption should proceed gradually, starting with hybrid systems where machine learning assists but does not replace existing rule-based handover algorithms. Such hybrid approaches could provide the benefits of improved prediction while still maintaining the safety and reliability of established mechanisms. Finally, the broader societal impact of deploying AI/ML in networks relates to trust. Users must trust that their connectivity will be reliable and that machine learning will not degrade their experience. Operators must trust that models will not introduce unexpected behaviors that compromise network stability. Regulators may also demand assurances of fairness, transparency, and accountability. The outcomes of this thesis therefore emphasize not only the technical promise of AI/ML in mobility management, but also the need for careful validation, ethical oversight, and transparent communication.

6 Conclusion

6.1 Summary of Aim and Research Questions

The aim of this thesis was to investigate the application of machine learning methods to predict mobility handovers in 5G networks, focusing on both L3 handover and L1/L2-triggered mobility. Two algorithms, Random Forest and XGBoost, were evaluated across four different scenarios (1UE and 10UE, pedestrian and vehicular) using simulation data. The research questions addressed three main points: first, whether machine learning methods can accurately classify handover versus non-handover cases; second, how performance differs across mobility and load conditions; and third, what methodological challenges arise when applying machine learning in this domain. It should be noted that although the theoretical background discussed supervised, unsupervised, and reinforcement learning approaches, the experimental part of this thesis focused exclusively on supervised methods, specifically Random Forest and XGBoost, due to their robustness and suitability for the available mobility datasets.

The motivation behind these questions stems from the critical role that mobility management plays in modern cellular systems. A missed handover may cause service interruption or call drops, while unnecessary handovers increase signalling overhead and reduce efficiency. Understanding whether machine learning can predict these events better than traditional threshold-based approaches is therefore both scientifically and practically important.

6.2 Answers to Research Questions

RQ1: Can machine learning models accurately classify handover versus non-handover cases? The experiments demonstrated that both Random Forest and XGBoost are capable of separating handover and non-handover events with very high accuracy, often reaching values close to 100%. However, accuracy alone was misleading in several cases, particularly in the 1UE scenarios. For Random Forest in the 1UE pedestrian case, accuracy was 0.99, but this came from predicting nearly all cases as handovers. Recall was perfect (1.0), but precision collapsed to 0.08, yielding a very poor F1 0.15. XGBoost in the same scenario behaved in the opposite way: it avoided false positives almost entirely, while still reporting high accuracy. These results show that accuracy cannot be used in isolation, and that precision, recall, and F1 are essential to judge model quality. In richer datasets, such as the 10UE vehicular

scenario, both models achieved accuracy close to 100%, recall 0.99–1.0, precision 0.91, and F1 0.95, showing that machine learning can classify mobility events almost perfectly under realistic conditions.

RQ2: How does performance differ across mobility and load conditions? The results confirmed that dataset richness and separability strongly influenced performance. In pedestrian scenarios, Random Forest always favored recall, never missing a handover, but producing excessive false positives when data was sparse. Precision improved from 0.08 in 1UE pedestrian to 0.36–0.38 in 10UE pedestrian, thanks to more consistent handover patterns. XGBoost, in contrast, was conservative: in 1UE pedestrian it predicted fewer handovers, but in 10UE pedestrian recall improved to 0.60–0.73 while precision remained high at 0.79. In vehicular scenarios, both models improved further. Random Forest maintained perfect recall and achieved high precision (0.91) in 10UE vehicle, while XGBoost achieved a similar balance, with recall 0.99 and precision 0.91. The models thus converged in performance when datasets were both rich and separable, while in sparse pedestrian cases their tendencies diverged dramatically.

RQ3: What methodological challenges arise when applying machine learning to mobility management? The study revealed two main challenges: dataset imbalance and dataset separability. In 1UE scenarios, imbalance led to misleadingly high accuracy but poor precision or recall. In the 1UE vehicle case, separability was trivial, allowing near-perfect classification without meaningful learning. These cases show that simulation-based datasets can inflate performance. The 10UE scenarios provided more realistic tests, exposing meaningful trade-offs between recall and precision, and showing how parameter tuning (especially XGBoost's learning rate) influenced results. These findings emphasize that accuracy alone is not sufficient, that careful dataset design is critical, and that hyperparameter sensitivity differs across algorithms.

6.3 Implications

From a technical perspective, the results demonstrate that Random Forest and XGBoost are both viable for mobility prediction, but with different trade-offs. Random Forest is robust, stable, and strongly recall-oriented, making it suitable for scenarios where reliability is more important than efficiency. XGBoost is more flexible and can achieve better precision–recall balance, but only when tuned carefully, particularly with respect to learning rate. In practice, this suggests that Random Forest could be used to guarantee service continuity, while XGBoost could reduce unnecessary handovers in denser networks.

From an ethical standpoint, the use of machine learning in mobility management raises questions of transparency and accountability. Ensemble methods, including Random Forest and XGBoost, are not easily interpretable, which complicates explaining decisions to operators and regulators. If a handover is missed or triggered unnecessarily, accountability must be clear. Moreover, fairness must be considered: a model trained primarily on vehicular scenarios may underperform for pedestrians, leading to unequal user experiences. Future deployments should therefore include interpretability tools, fairness analysis, and safeguards to ensure accountability.

From a societal and industrial perspective, the results show both promise and caution. Reliable and efficient handovers directly affect user experience, from maintaining video calls to ensuring safety in connected vehicles. The telecom industry can integrate machine learning into network management, but only gradually, starting with hybrid systems where AI supports but does not replace traditional algorithms. Such approaches could leverage the strengths of machine learning while retaining the safety and reliability of established methods. Ultimately, user trust and regulatory approval will depend on transparency, fairness, and demonstrable improvements in quality of service.

6.4 Future Work

Several directions for future research follow from this study. First, the models should be tested on real-world measurement data to ensure external validity, as simulation-based datasets sometimes exaggerate separability or imbalance. Second, feature engineering could be expanded to include SINR, beam quality, interference levels, or traffic load, which may improve prediction in complex scenarios. Third, interpretability methods such as SHAP or permutation importance should be applied to explain model behavior, increasing operator trust. Fourth, fairness analysis should be carried out to ensure balanced performance across different user types and mobility patterns. Finally, reinforcement learning could be explored as a more adaptive approach compared to static classifiers, allowing handover strategies to evolve dynamically with network conditions. Together, these directions would help bridge the gap between experimental studies and operational deployment of AI/ML in future mobile networks. While this thesis focused on supervised learning methods (Random Forest and XGBoost), reinforcement learning remains a promising direction for future studies. RL could enable more adaptive and autonomous handover management by allowing models to learn policies directly through interaction with dynamic network environments.

In summary, this thesis has provided both technical evidence and critical reflection on the use of AI/ML for mobility management. While challenges remain, the results contribute to ongoing efforts to design reliable, efficient, and transparent mobility solutions in 5G and beyond. At the same time, they lay the groundwork for future research that explores more adaptive methods, such as reinforcement learning, and validates these approaches on real-world datasets. Together, these developments will be essential in building the next generation of intelligent, trustworthy, and future-proof mobility management systems.

Bibliography

[1] Federica Rinaldi, Alessandro Raschellà, and Sara Pizzi. "5G NR system design: a concise survey of key features and capabilities". In: *Wireless Networks* 27.8 (2021), pp. 5173–5188.

[2] Jiyong Pang, Shaobo Wang, Zhenfei Tang, Yanmin Qin, Xiaofeng Tao, Xiaohu You, and Jinkang Zhu. "A new 5G radio evolution towards 5G-Advanced". In: *Science China Information Sciences* 65.9 (2022), 191301:1–191301:45.

[3] E. Dahlman, S. Parkvall, and J. Skold. *5G NR: The Next Generation Wireless Access Technology*. Academic Press, 2018.

[4] S. Hailu, M. Saily, and O. Tirkkonen. "RRC State Handling for 5G". In: *IEEE Communications Magazine* 57.1 (2019), pp. 106–113.

[5] Zhiyong Li, Xiaoyan Tang, Binbin Zheng, Wen Pan, Shaohuai Zheng, and Yiwen Shen. "Synchronization signals design and detection for 5G and beyond wireless networks: a review". In: *EURASIP Journal on Wireless Communications and Networking* 2023.1 (2023), pp. 1–30.

[6] Muhammad Tayyab, Xavier Gelabert, and Riku Jäntti. "A Survey on Handover Management: From LTE to NR". In: *IEEE Access* 7 (2019), pp. 118907–118930.

[7] Changsung Lee, Jaewook Jung, and Jong-Moon Chung. "Intelligent dual active protocol stack handover based on double DQN deep reinforcement learning for 5G mmWave networks". In: *IEEE Transactions on Vehicular Technology* 71.7 (2022), pp. 7572–7584.

[8] Jędrzej Stańczak, Umur Karabulut, and Ahmad Awada. "Conditional handover in 5G—Principles, future use cases and FR2 performance". In: *2022 International Wireless Communications and Mobile Computing (IWCMC)*. IEEE. 2022, pp. 660–665.

[9] Christopher M. Bishop. *Pattern Recognition and Machine Learning*. Springer, 2006.

[10] Leo Breiman. "Random forests". In: *Machine Learning* 45.1 (2001), pp. 5–32.

[11] Tianqi Chen and Carlos Guestrin. "XGBoost: A scalable tree boosting system". In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 2016, pp. 785–794.

[12] Corinna Cortes and Vladimir Vapnik. "Support-vector networks". In: *Machine Learning* 20.3 (1995), pp. 273–297.

[13] Ian T. Jolliffe. *Principal Component Analysis*. 2nd ed. Springer-Verlag, 2002.

[14] Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, 2018.

[15] Z. Ali, N. Baldo, J. Mangues-Bafalluy, and L. Giupponi. "Machine learning based handover management for improved QoE in LTE". In: *2016 IEEE/IFIP Network Operations and Management Symposium (NOMS)*. 2016, pp. 794–798.

[16] M. F. Ahmad Fauzi, R. Nordin, N. F. Abdullah, and H. A. H. Alobaidy. "Mobile Network Coverage Prediction Based on Supervised Machine Learning Algorithms". In: *IEEE Access* 10 (2022), pp. 55782–55793.

[17] S. B. Iqbal, U. Karabulut, A. Awada, A. N. Barreto, P. Schulz, and G. P. Fettweis. "RACH Optimization with Decision Tree Based Supervised Learning for Conditional Handover in 5G Beamformed Systems". In: *arXiv preprint arXiv:2309.09840* (2023).

[18] T. Cabral de Brito Guerra, Y. R. Dantas, and V. A. Sousa Jr. "A Machine Learning Approach for Handover in LTE Networks with Signal Obstructions". In: *Journal of Communication and Information Systems* 35.1 (2020), pp. 271–289.

[19] Z. Wang, L. Li, Y. Xu, H. Tian, and S. Cui. "Handover Control in Wireless Systems via Asynchronous Multiuser Deep Reinforcement Learning". In: *IEEE Internet of Things Journal* 5.6 (2018), pp. 4296–4307.

[20] J. Tanveer, A. Haider, R. Ali, and A. Kim. "An Overview of Reinforcement Learning Algorithms for Handover Management in 5G Ultra-Dense Small Cell Networks". In: *Applied Sciences* 12.1 (2022), p. 426.

[21] W. Qin, Y. Teng, M. Song, Y. Zhang, and X. Wang. "AQ-learning approach for mobility robustness optimization in LTE-SON". In: *2013 15th IEEE International Conference on Communication Technology*. 2013, pp. 818–822.

[22] M. K. Dahouda, S. Jin, and I. Joe. "Machine learning-based solutions for handover decisions in non-terrestrial networks". In: *Electronics* 12.8 (2023), p. 1759.

[23] M. E. Morocho-Cayamcela, H. Lee, and W. Lim. "Machine Learning for 5G/B5G Mobile and Wireless Communications: Potential, Limitations, and Future Directions". In: *IEEE Access* 7 (2019), pp. 137184–137206.

[24] Q. Xue, J. Guo, B. Zhou, Y. Xu, Z. Li, and S. Ma. "AI/ML for Beam Management in 5G-Advanced: A Standardization Perspective". In: *IEEE Vehicular Technology Magazine* 19.4 (2024), pp. 64–72.

[25] N. Jayaweera, A. Bonfante, M. Schamberger, A. M. A. Tehrani, T. Sanguanpuak, P. Tilak, K. Jayasinghe, F. W. Vook, and N. Rajatheva. "5G-Advanced AI/ML Beam Management: Performance Evaluation with Integrated ML Models". In: *arXiv preprint arXiv:2404.15326* (2024).